TAMID QUANT — STRATEGY ONE PAGER

Bitcoin Digital Options: Kalshi Price vs. Model Probability

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The Idea

Kalshi lists hourly "Bitcoin above $K at 5pm ET" contracts that pay $1 if true. Each is a digital option, so its fair price is the probability that BTC finishes above K. We estimate that probability from a volatility forecast, trade the contract when it is more than 4c away from the model, and optionally hedge the BTC delta. Tested on 19,299 contracts over 816 expiries (Jul 2024 to Nov 2026): the market is already well priced, and no edge survives costs. The one robust pattern, overpriced tails, is the basis of the separate tail strategy one-pager.

Mathematical Framework

Core Model: *Digital pricing.*

With S the spot price, σ the forecast volatility and h the hours to expiry, $p = P(S_T > K) = N(d_2)$, $d_2 = [\ln(S/K) - \sigma^2 h/2] / (\sigma\sqrt{h})$. The hedge ratio is the delta $\Delta = \varphi(d_2) / (S \cdot \sigma\sqrt{h})$.

Signal Construction: *Edge = model p − Kalshi price, using only data up to the signal time:*

σ^2 from an EWMA of hourly returns ($\alpha = 0.03$); GARCH(1,1) and 168-hour realized vol as alternatives. Tails from filtered historical simulation: the empirical distribution of the past 90 days of standardized returns, kernel-smoothed, replacing the normal. Quality check: Brier score of model vs. market price.

Entry / Exit Logic: *Signal 6 hours before expiry, fill one hour later at the Kalshi bid or ask:*

If p − ask > 4c, buy YES; if bid − p > 4c, sell YES. Hold to settlement. The hedged variant also holds $-(\text{side}) \cdot \Delta$ in BTC, rebalanced hourly at a 5bps cost on turnover. Strikes within ±6% of spot, at most 30 contracts per expiry.

Position Sizing: *One contract per qualifying strike.*

Kalshi collateral is at most $1 per contract, and the hedge holds about $46 of BTC per $1 contract.

Key Assumptions

- Short-horizon BTC volatility and tails resemble the past 90 days
- Hourly-candle bid/ask quotes are fillable, with fee $0.07 \cdot P \cdot (1 - P)$ (no depth data)
- Each expiry day is one draw: 504 trading days, not 1,918 trades. In-sample before Dec 20 2025, out-of-sample after

Key Risks

- No pricing edge: Brier score is 0.0584 for the market and 0.0583 for the model (difference +1.2e-4, 95% CI −4.8 to +7.1 e-4). Both GARCH variants are significantly worse than the market
- Unhedged +0.80c per contract over 1,918 trades is not significant. 12h loses at every threshold, 3h loses at 2-6c, and fees take 1.0c of the 1.8c gross edge (double the fee and it loses)
- The hedge cuts the standard deviation from 34c to 25c but costs 4.6c per contract (−4.19c net, still −1.43c at 2bps)
- The model is better calibrated than the market in the tails: contracts priced 2-5c settled YES 0.8% (model 1.5%), and 95-98c settled 98.6%. That longshot bias is the only robust pattern

At a Glance

Asset Class:	*Crypto / Prediction Markets*
Strategy Type:	*Model vs. Market Mispricing*
Direction:	*Long/Short Digitals (+ Optional BTC Hedge)*
Holding Period:	*Hours (Hold to Expiry)*
Universe:	*Kalshi KXBTCD vs. Coinbase BTC-USD*

Backtest Results (816 expiries)

Variant (6h, edge > 4c)	Net / contract	95% CI
Unhedged, all	+0.80c	−1.2c to +2.7c
Unhedged, out-of-sample	+1.16c	−1.4c to +3.9c
BTC-hedged (5bps)	−4.19c	−5.6c to −2.8c
Unhedged, fee x2	−0.17c	

1,918 trades over 504 days, Jul 2024 to Nov 2026, net of fees.

Data Requirements

- Kalshi KXBTCD hourly bid, ask, volume and settlement - public API
- Coinbase BTC-USD hourly candles for volatility and the hedge - public API
- 19,299 settled contracts, 816 expiries (5pm ET), 266,501 hourly quotes
- Order-book depth (not yet collected)

Key Parameters (starting values)

- Edge threshold: 4c (tested 2c to 10c)
- Signal horizon: 6h before expiry (tested 3h, 12h)
- Volatility: EWMA $\alpha = 0.03$; FHS 90-day window
- Hedge cost: 5bps (tested 0 to 10bps); fee $0.07 \cdot P \cdot (1 - P)$

Open Questions

- Do contracts misprice after sharp BTC moves, when the quote lags spot? Needs minute-level data
- Can resting (maker) orders avoid the spread and fee that erased the edge?
- Is there a Kalshi vs. Polymarket gap for BTC?